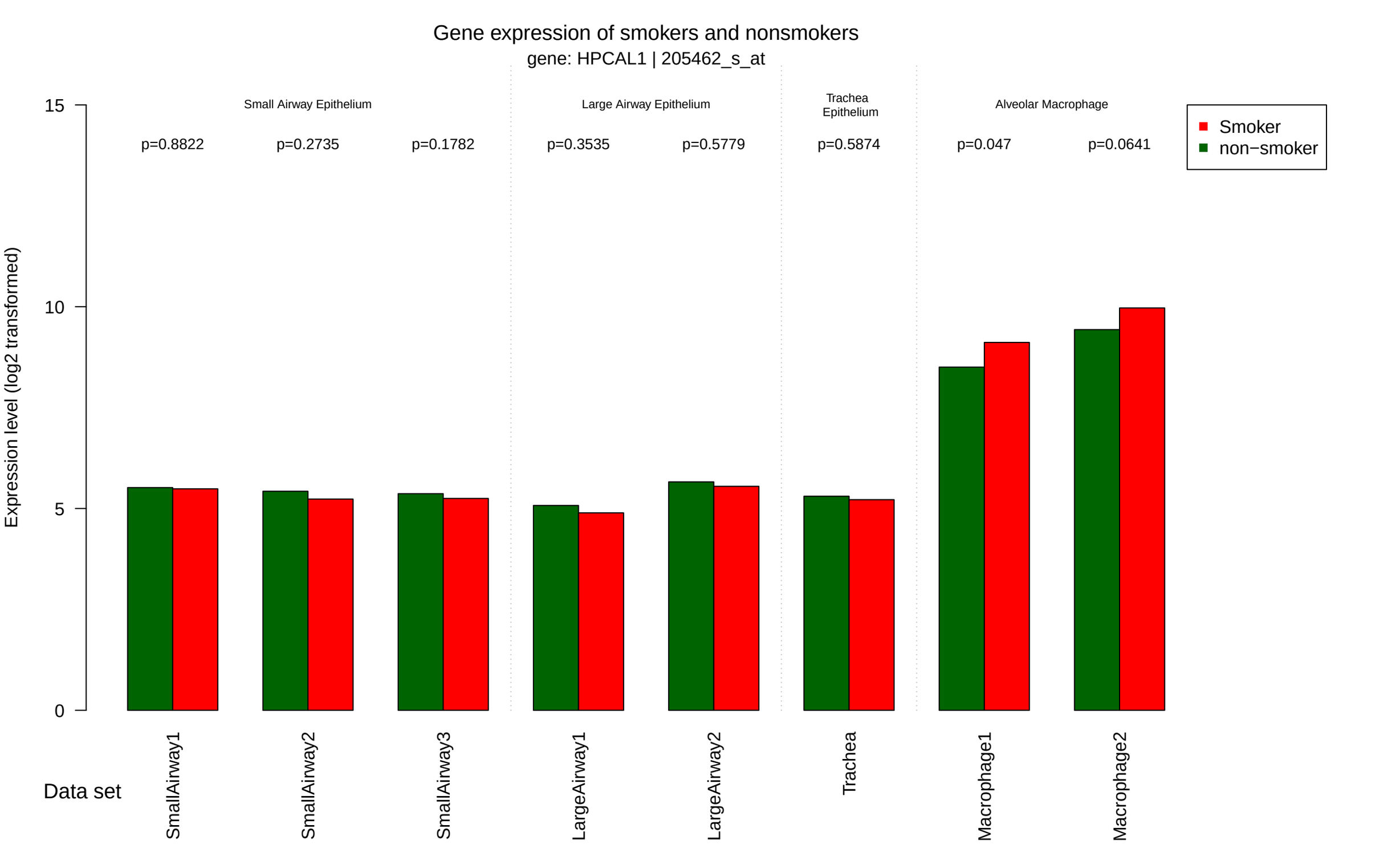
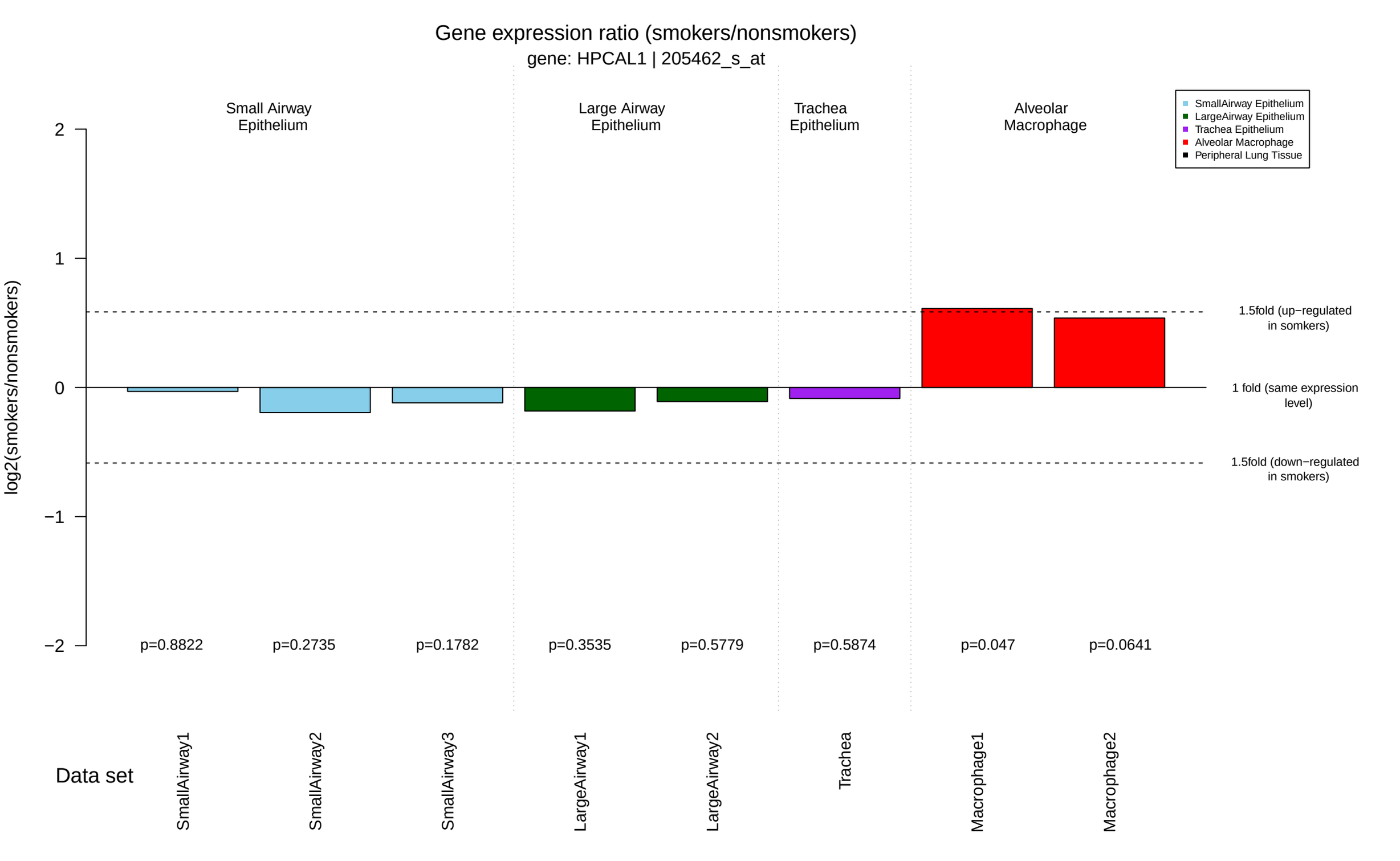
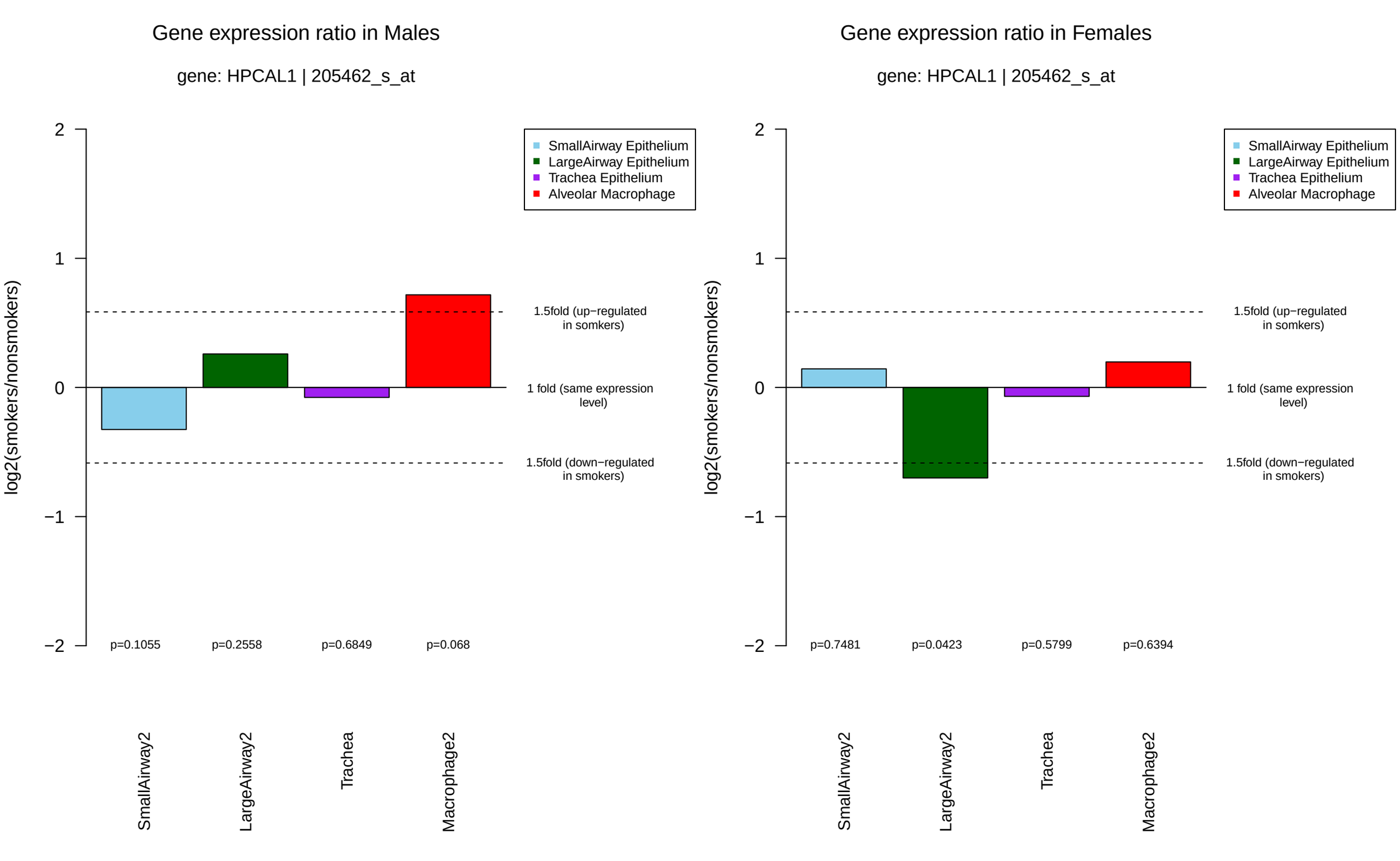
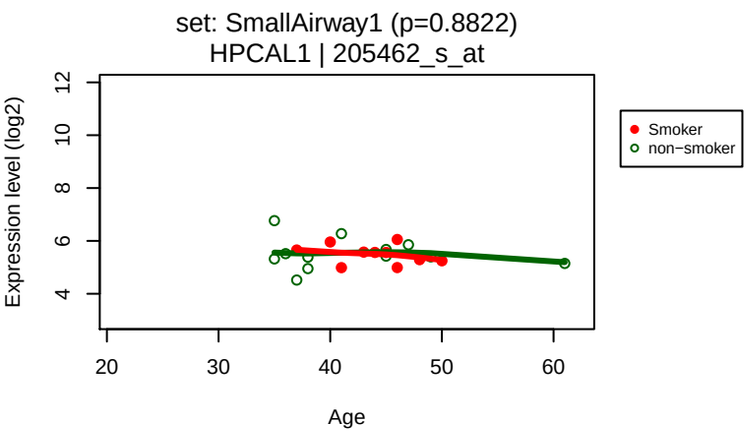
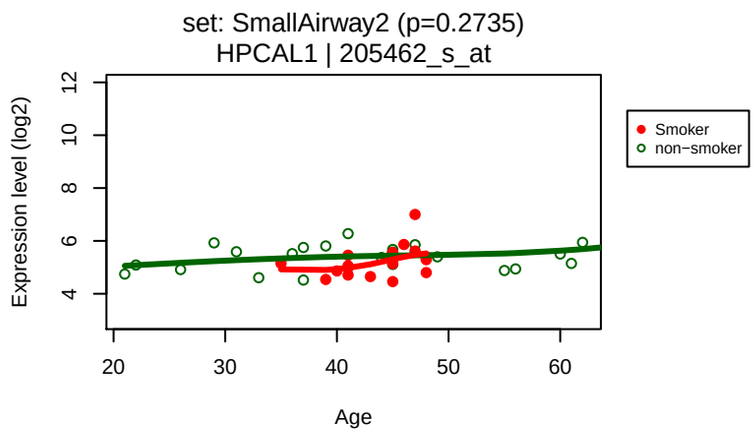
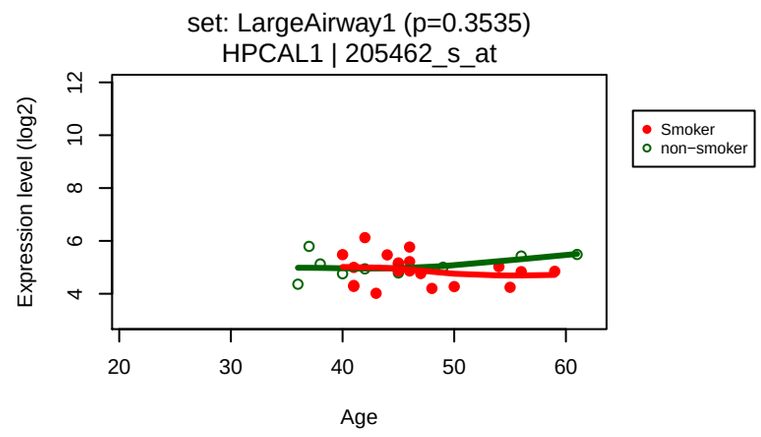

Gene expression of smokers and nonsmokers
gene: HPCAL1 | 205462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8822
p=0.2735
p=0.1782
p=0.3535
p=0.5779
p=0.5874
p=0.047
p=0.0641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

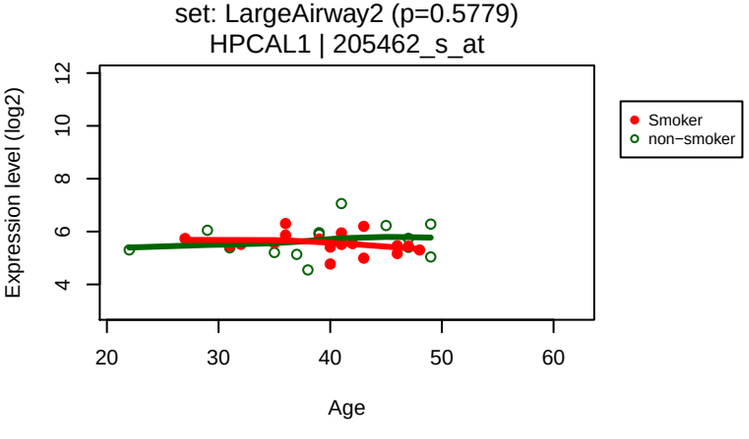

Gene expression ratio (smokers/nonsmokers)
gene: HPCAL1 | 205462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8822
p=0.2735
p=0.1782
p=0.3535
p=0.5779
p=0.5874
p=0.047
p=0.0641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

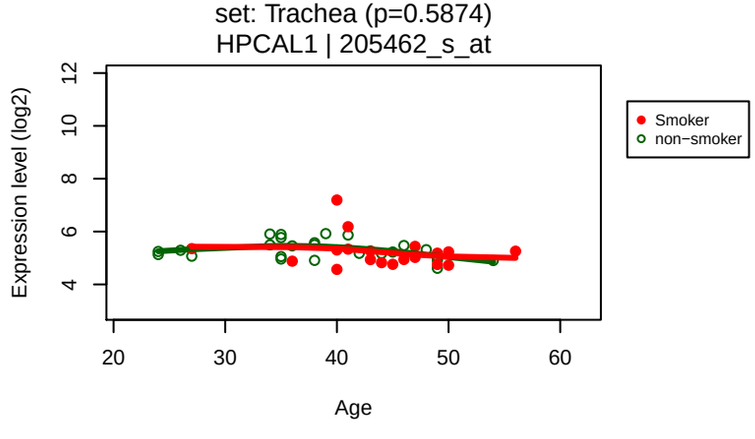

Gene expression ratio in Males
gene: HPCAL1 | 205462_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1055
p=0.2558
p=0.6849
p=0.068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPCAL1 | 205462_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7481
p=0.0423
p=0.5799
p=0.6394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

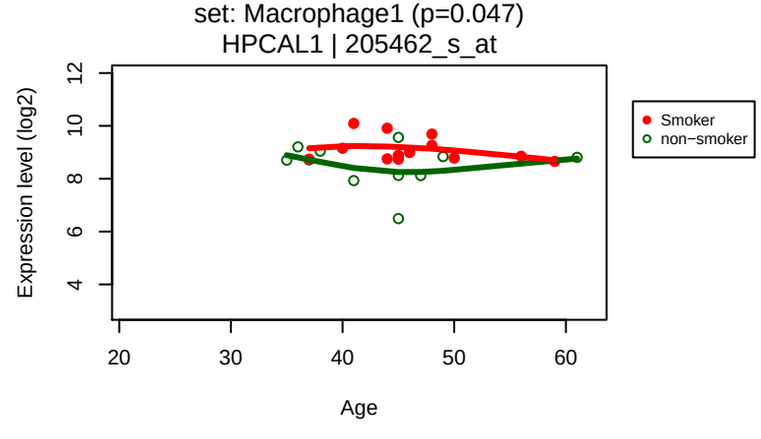

set: SmallAirway1 (p=0.8822)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker

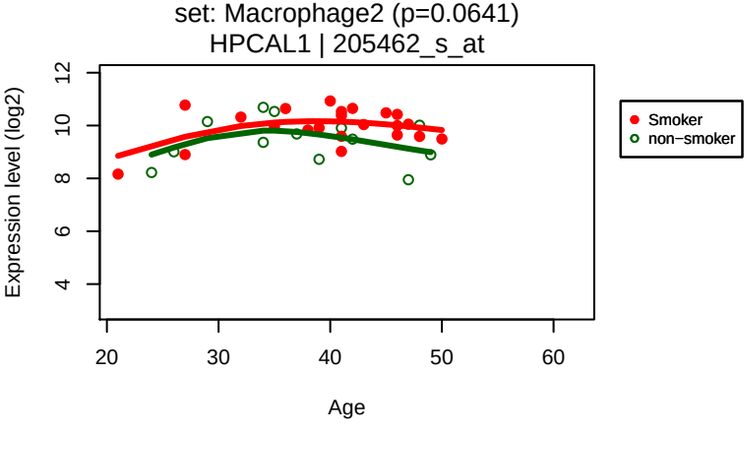

set: SmallAirway2 (p=0.2735)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker

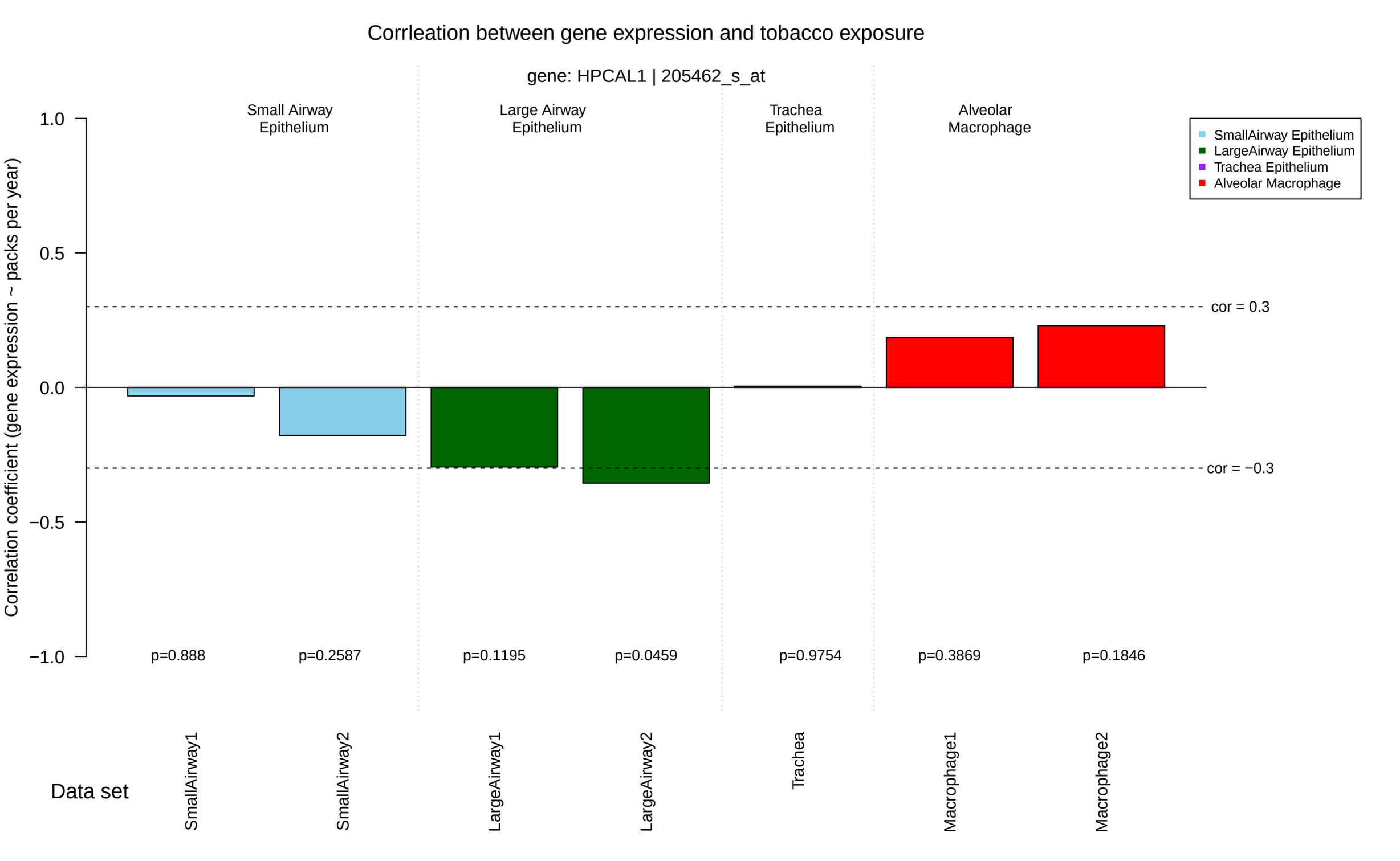

set: LargeAirway1 (p=0.3535)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5779)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5874)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.047)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0641)
HPCAL1 | 205462_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPCAL1 | 205462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.888
p=0.2587
p=0.1195
p=0.0459
p=0.9754
p=0.3869
p=0.1846
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2